

Even Better Privacy

Post-quantum identity and secure messaging

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Abstract

Even Better Privacy (EBP) is a post-quantum identity and secure communications protocol. Each *identity* pairs signing with a separate encryption and key-encapsulation role. Confidentiality uses ML-KEM (FIPS 203); authentication and integrity use ML-DSA or SLH-DSA (FIPS 204 and FIPS 205); bulk data are protected with AES-GCM after encapsulation. These choices follow NIST-approved post-quantum standards for the asymmetric layer. This document defines the protocol and its components.

1 Identity

EBP defines an identity as a pair of public keys: one for signing and the other for encryption.

1.1 Signing Key

Options for the signing key are:

- ML-DSA (Dilithium)
- SLH-DSA (SPHINCS+)

Both of these options are great for separate purposes. Later in this paper we discuss how to use them together in a hierarchical relationship to benefit from the strengths of both.

1.2 Encryption Key

The encryption key is always ML-KEM (Kyber) because that is the only NIST approved post-quantum KEM. We will consider adding other KEMs upon approval from NIST.

1.3 Fingerprint

The fingerprint is a bech32-encoded merkle root of the two public keys:

- **Left leaf:** SHA-256(base64-decoded signing public key bytes)
- **Right leaf:** SHA-256 of the encryption public key decoded from its lowercase hexadecimal string (not the UTF-8 bytes of the hex digits)

The merkle tree design means the fingerprint can be verified without both keys present. This is useful in bandwidth-constrained scenarios.